

Q1. Smart Ambulance Navigation — A* Search

Use A* because road costs and heuristic values are given.

Formula: $f(n) = g(n) + h(n)$

Start: $S \rightarrow A$, cost = 2

$g(A)=2$, $h(A)=7$, so $f(A)=9$.

Expand A:

B: $g=6$, $h=6$, $f=12$

D: $g=8$, $h=6$, $f=14$

E: $g=6$, $h=3$, $f=9$

Choose E.

Expand E:

F: $g=10$, $h=1$, $f=11$. Choose F.

Expand F:

G: $g=12$, $h=0$, $f=12$. Goal reached.

Selected path: $S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$

Cost = $2+4+4+2 = 12$.

Another path: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

Cost = $2+4+3+2 = 11$.

Therefore the selected path is NOT optimal.

Reason: the given heuristic is not admissible everywhere.

Final: A* path cost = 12; true optimal cost = 11.

Q2. Autonomous Warehouse Robot — Online Search

The robot initially does not know all obstacles.

Suitable method: Online A* / LRTA*.

Steps:

1. Start at S.
2. Sense nearby cells.
3. Choose the best available move toward G.
4. Move one step.
5. If a new obstacle is found, mark it blocked.
6. Update the map and replan.
7. Continue until G is reached.

Example: if (4,2) is discovered as an obstacle,
the robot avoids it and searches another route, e.g.

$(5,1) \rightarrow (5,2) \rightarrow (5,3) \rightarrow (5,4) \rightarrow \dots \rightarrow G$

Online search: incomplete map, senses obstacles and replans.

Offline search: complete map is known before searching.

Conclusion: Online search is suitable for an unknown warehouse.

Q3. University Examination Scheduling — CSP

Variables: E1, E2, E3, E4, E5, E6

Domains: {T1, T2, T3, T4}

Constraints:

$E1 \neq E2, E1 \neq E3, E2 \neq E4,$

$E3 \neq E5, E4 \neq E6, E2 \neq E5.$

Hall capacities: A=120, B=90, C=60.

Students: E1=100, E2=80, E3=70, E4=60, E5=90, E6=50.

MRV + Forward Checking:

$E1=T1 \rightarrow E2, E3$ remove T1.

$E2=T2 \rightarrow E4, E5$ remove T2.

$E3=T2 \rightarrow E5$ cannot use T2.

$E4=T1 \rightarrow E6$ cannot use T1.

$E5=T3.$

$E6=T2.$

Final timetable:

$E1 \rightarrow T1 \rightarrow \text{Hall A} \rightarrow 100$

$E2 \rightarrow T2 \rightarrow \text{Hall B} \rightarrow 80$

$E3 \rightarrow T2 \rightarrow \text{Hall A} \rightarrow 70$

$E4 \rightarrow T1 \rightarrow \text{Hall C} \rightarrow 60$

$E5 \rightarrow T3 \rightarrow \text{Hall A} \rightarrow 90$

$E6 \rightarrow T2 \rightarrow \text{Hall C} \rightarrow 50$

All student-conflict constraints are satisfied.

All hall capacities are satisfied.

Therefore the timetable is VALID.

Q4. Game-Playing AI — Minimax & Alpha-Beta

Minimax:

Left MIN: $\max(3,5,6)=6$; $\max(9,1,2)=9$; $\max(0,-1,4)=4$.

Left MIN = $\min(6,9,4)=4$.

Middle MIN:

$\max(7,5,8)=8$; $\max(6,2,3)=6$; $\max(4,0,-2)=4$.

Middle MIN = 4.

Right MIN:

$\max(1,2,3)=3$; $\max(5,-1,2)=5$; $\max(6,4,7)=7$.

Right MIN = 3.

Root MAX = $\max(4,4,3) = 4$.

Best value = 4. Choose left or middle branch.

Alpha-Beta pruning:

At the left branch, after seeing $9 \geq 6$, values 1 and 2 are pruned.

At the right branch, first value gives 3.

Root already has $\alpha=4$, so $\beta=3 \leq \alpha=4$.

The remaining two right-side branches are pruned.

Final value = 4.

Alpha-Beta gives the same answer with fewer evaluations.

Q5. Delivery Route Optimization — Hill Climbing

Hill climbing is a local-search optimization method.

For minimization, move to a neighbour with smaller cost.

Starting cost = 23 km.

Best neighbour = 22 km.

Since $22 < 23$, move to the better neighbour.

23 km $\rightarrow$ 22 km

At 22 km, all neighbours have cost ≥ 22 km.

Therefore 22 km is a local optimum.

Is it a global optimum?

Not necessarily. Hill climbing checks only neighbouring solutions.

To escape a local optimum:

1. Random Restart Hill Climbing.
2. Simulated Annealing.
3. Stochastic Hill Climbing.

Conclusion: Final local solution = 22 km.

Use random restart or simulated annealing to search for a better global solution.